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Neuroprotective properties of *Cantharellus cibarius* polysaccharide fractions in different *in vitro* models of neurodegeneration

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ABSTRACT

The percentage of people suffering from neurodegenerative diseases is constantly increasing, because of that searching for substances able to prevent or inhibit neuronal death sseems to be reasonable. Because of the high popularity the search of new neuroprotective agents we started from *Cantharellus cibarius*. Neuroprotective properties of *C. cibarius* polysaccharides fractions was investigated in different models of neurodegeneration including trophic stress, excitotoxicity and andoxidative stress. Fractions influence on neurons viability was examined using Neurite Outgrowth Staining, MTT and LDH tests, while antioxidant capacity was determined by commercial antioxidant assays. Performed studies revealed beneficial effect of *C. cibarius* fractions (CC2a, CC3) on neurons viability and neurite outgrowth in normal and different stress conditions. Both tested fractions have shown antioxidant capacity and effectively neutralize the negative changes induced by glutamatergic system activators. Discovered neuroprotective properties of investigated compounds suggested the their use for developing effective and safety therapeutic strategy for neurodegenerative diseases.

1. Introduction

Despite the progress that has been made in neuroscience in recent years, the percentage of people suffering from neurodegenerative diseases like amyotrophic lateral sclerosis, Alzheimer's, Huntington's, Parkinson's is constantly increasing (Brown, Lockwood, & Sonawane, 2005). Due to the insufficient effectiveness of the available therapeutic strategies, high hopes are placed in neuroprotection defined as prevention of neuronal structure and/or function from various damaging factors using substances capable of preventing or inhibiting neuronal death (Brown et al., 2005; Vajda, 2002).

Due to the wide spectrum of biological activities and the large therapeutic potential of polysaccharides isolated from *Basidiomycetes* (Devi & Maiti, 2016; Lemieszek & Rzeski, 2012; Wasser, 2002; Zaidman, Yassin, Mahajna, & Wasser, 2005) we decided to take a look at this group of compounds aiming to discover effective and safe neuroprotective agents. So far neuroprotective abilities have been shown in case of polysaccharides isolated from *Agaricus bisporus* (Amal, Mona, & Rasha, 2015), *Cordyceps cicadae* (Olatunji et al., 2016), *Cordyceps militaris* (Tsai, Yen, & Yang, 2015), *Ganoderma lucidum* (Aguirre-Moreno et al., 2013; Gokce et al., 2015; Lai et al., 2008; Tello et al., 2013; Zhang et al., 2011; Zhao, Lin, Liu, & Lin, 2004; Zhou et al., 2010), *Hericium erinaceus* (Cheng, Tsai, Lien, Lee, & Sheu, 2016; Han, Ye, & Wang, 2013; Park et al., 2002; Wong, Kanagasabapathy, Bakar, Phan, & Sabaratnam, 2015), *Innonotus obliquus* (Yue, Xiuhong, Shuyan, & Zhonghua, 2015), *Lignosus rhinocerotis* (Seow et al., 2015), *Sparassis crispa* (Hu et al., 2016), *Tremella fuciformis* (Jin, Hu, Zhang, & Liu, 2016; Kim et al., 2007). Despite the limited evidences, which proved neuroprotective effect of mushrooms polysaccharides, available data indicate such potential and provide incentives for further research.

Because of the high popularity both in the terms of its occurrence (cosmopolitan species) as well as consumption (species prized for its taste and smell) the search of new neuroprotective agents we started from *Cantharellus cibarius* (golden chanterelle). Furthermore, recent reports which gradually discover beneficial properties of *C. cibarius* including anticancer, anti-inflammatory, immunomodulatory, antigenotoxic, antiaging and antioxidant (Drewnowska & Falandysz, 2015; Ebrahimzadeh, Nabavi, Nabavi, & Eslami, 2010; Ebrahimzadeh,

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Safdari, & Khalili, 2015; Hong et al., 2012; Muszyńska, Sułkowska-Ziaja, & Ekiert, 2011; Muszyńska, Kała, Firlej, & Sułkowska-Ziaja, 2016; Palacios et al., 2011; Valentão et al., 2005) attracted our attention to bioactive compounds presented in golden chanterelle as well as pointed out the need for further investigation to broaden the therapeutic applications of this mushroom.

The main aim of the presented study was the evaluation the neuroprotective activity of polysaccharide fractions isolated from *C. cibarius* in different *in vitro* models of neurodegeneration, including trophic stress, excitotoxicity and oxidative stress. Studies were conducted in neurons derived from human neuroblastoma cell line SH-SY5Y.

2. Materials and methods

2.1. Reagents

All reagents and kits were purchased from Sigma-Aldrich (St Louis, MO, USA) unless otherwise indicated. Stock solutions of polysaccharide fractions (10 mg/ml) were prepared in PBS (buffered saline solution) and stored at 4 °C. Working solutions (10, 25, 50, 100 µg/ml) were prepared by dissolving an appropriate stock solution in culture medium.

2.2. Fungal origin

The wild-grown fruiting bodies of *Cantharellus cibarius* were collected in a chestnut orchard (Macedo de Cavaleiros 41°35' N and 6°57' W, 700 m altitude) and in a mixed forest stand of pine and oak (Vila Real (41°19' N and 7°44' W, 479 m altitude). Representative voucher samples of the wild-gathered species were deposited at the mycological herbarium of the University of Trás-os-Montes e Alto Douro.

2.3. Extraction and purification of Cantharellus cibarius fractions

For the preparation of the AIR (Alcohol Insoluble Residue), *Cantharellus cibarius* mushrooms were freeze-dried and ground to a fine powder. To 700 ml of 80% ethanol at 80 °C 100 g of mushroom powder were added, and the suspension was boiled with stirring during 1 h. After this time, the insoluble material was filtered through glass fiber and washed twice with 50 ml of 80% ethanol. The AIR was dried in a forced air oven at 50 °C during 12 h. For the preparation of the water soluble material (WSM), to 700 ml of boiling water, it was added 100 g of the AIR, and the suspension was boiled with stirring during 1 h. After this time, the insoluble material was separated from the supernatant by centrifugation (6000 rpm, 20 min, 4 °C). The water insoluble material was washed with water and separated again by centrifugation as before. The supernatant were combined, dialyzed (MW cut off 12–14 kDa) against water (6 water renewals) and freeze dried yielding the *Cantharellus cibarius* yielding the water soluble biopolymers (WSB). The *Cantharellus cibarius* WSB, including polysaccharide and glycoproteins present in the WSM and isolated by dialysis were further purified by fractionation by anion exchange chromatography on a Q-Sepharose FF stationary phase (loaded on a XK16/20 column, Pharmacia). The eluent was a pH 4.5 Na-acetate 5 mM buffer containing 0.02% sodium aside. WSB (at a concentration of 1 mg/ml) was applied to the column and, after application, the column was flushed with a minimum of 4 column volumes of the initial buffer or until the absorbance at 280 nm reached the initial level. The retained material was eluted by gradient elution, 0 to 500 mM NaCl in 5 h, 500 mM to 1000 mM in 3 h and 1000 mM to 2000 mM in 2 h. Fractions (2 ml) were collected and assayed for sugars by the phenol–sulfuric acid method and continuously monitored at 280 nm. The appropriated fractions were polled, dialysed (12–14 kDa cut-off), and freeze-dried. Four fractions were obtained one not retained on the anion-exchange column CC1 and three retained on the anion exchange column and eluted with increasing salt concentrations CC2a, CC2b, CC3. Sugar composition of the AIR, WSM, WSB and purified fractions were determined after hydrolysis with sulfuric acid 1 M at 100 °C during 2.5 h, using between 2 to 5 mg of sample. After hydrolysis 0.5 ml of internal standard was added (2-deoxyglucose at 1 mg/ml), and the solution was diluted 10 times before analysis by high performance anion-exchange chromatography with pulsed amperometric detection (HPAEC-PAD, ICS-3000, Dionex). The separation was performed with a CarboPac PA-20 column (150 mm × 3 mm) with a CarboPac PA20 pre-column (Dionex) using eluent A – 1.25 mM NaOH solution containing 2 mM $Ba(OH)_2$, eluent B – 400 mM sodium acetate containing 2 mM $Ba(OH)_2$ and eluent C – 500 mM NaOH containing 2 mM $Ba(OH)_2$. The eluent was kept under nitrogen all times to reduce carbonate buildup and biological contamination. The injection volume was 5 µl, the flow rate was 0.3 ml/min and the column temperature was maintained at 35 °C during the run. The following elution program was used: 0–19 min, 100% A, increase to 50% B until 27 min and maintained until 37 min; increase to 40% C and decreasing to 0% B until 47 min and maintained until 57 min. The column was conditioned with 100% A during 15 min before injection. Electrochemical detector consisted of Au working electrode, Ag/AgCl reference electrode, and Ti counter electrode. The ED cell waveform was +0.1 V from 0.00 to 0.40 s, then –2.0 V from 0.41 to 0.42 s, and a ramp –2.0 to +0.6 V from 0.42 to 0.43 s, followed by –0.1 V from 0.44 to 0.50 s (end of cycle). The integration region was from 0.2 s to 0.4 s.

2.4. Size exclusion chromatography

Size exclusion chromatography on Sephacryl S-400 HR was performed on a column width of 2.6 and 70 cm length (XK 26/70, Pharmacia) at a flow rate of 0.3 ml/min. Samples (2 mg) were suspended in 1 ml of 100 mM Na phosphate buffer, pH 6.5, containing 3 M urea. The same phosphate/urea buffer was used as the eluent. Fractions (1 ml) were collected and assayed for sugars with the phenol–H_2SO_4 method, and the eluent was continuously monitored at 280 nm. Exclusion and total volumes were calibrated with Blue Dextran and Glc, respectively.

2.5. Methylation analysis

Polysaccharides were activated with powdered NaOH and methylated with CH_3I (Ciucanu & Kerek, 1984; Isogai, Ishizu, & Nakano, 1985) as described by Coimbra, Delgadillo, Waldron, and Selvendran (1996), followed by a remethylation to ensure complete methylation of the polysaccharides (Nunes & Coimbra, 2001). The remethylated material was hydrolyzed with 2 M trifluoroacetic acid (Harris, Henry, Blackeney, & Stone, 1984), and the partially methylated sugars were reduced with NaBD4 and acetylated with acetic anhydride with 1-methylimidazole as catalyst. The partially methylated alditol acetates (PMAAs) were identified by gas chromatography-mass spectrometry (Nunes & Coimbra, 2001) and quantified by gas chromatography-flame ionization detection using the molar response factors of Sweet, Shapiro, and Albersheim (1975).

2.6. Cell line

Human undifferentiated neuroblastoma cell line SH-SY5Y was purchased from ECACC (European Collection of Cell Cultures, Salisbury, UK.). Cells were grown in 1:1 mixture of Ham's F12 nutrient and Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum (FBS), 1% non-essential amino acid solution, penicillin (100 U/ml) and streptomycin (100 mg/ml). Cells were maintained in a humidified atmosphere of 95% air and 5% CO2 at 37 °C (standard conditions).

2.6.1. Differentiation of SH-SY5Y towards neuronal cells

For induction of the neuronal differentiation, SH-SY5Y cells at a

density 1×10^6 cells/ml were resuspended in Alpha MEM Gluta MAX medium supplemented with 0.5 μM retinoic acid. Then cells were seeded on low attachment Petri dish and incubated for 5 days in standard conditions. The created neurospheres were collected and treated with 0.05% trypsin-EDTA for 15 min in 37 °C. After blocking trypsin with FBS cells were shaking in presence of 100 μg/ml DNAase I. Finally, the cell suspension was centrifuged at 1200 rpm for 5 min and resuspended in Neurobasal medium (Gibco) containing 2% B-27 supplement (Gibco), 2 mM L-glutamine, penicillin (100 U/ml) and streptomycin (100 mg/ml). The cells were seeded on 96-well microplates (Nunc) coated with poly-D-lysine at a density of 2×10^5 cells/mL and kept under standard conditions. The culture medium was changed on the 3rd day and 7th day. All experiments were carried out after 7 days of culture.

2.7. Cell viability assessment – MTT assay

Neurons cultured for 7 days were exposed to serial dilutions of tested fractions (10, 25, 50 i 100 μg/ml) used alone or in combination with 1 mM glutamate, 0.5 mM NMDA, 0.25 mM AMPA. Solutions of investigated compounds were prepared in the Neurobasal medium supplemented with 2% B-27 supplement, 2 mM L-glutamine, penicillin (100 U/ml) and streptomycin (100 mg/ml). In case of experiments performed in conditions of trophic stress solutions of tested fractions were prepared in the medium deprived of B-27 supplement. After 48 h of cells exposure to investigated compounds MTT solution (5 mg/ml in PBS) was added for 3 h. Resultant crystals were solubilized overnight in SDS buffer pH 7.4 (10% SDS in 0.01 N HCl) and the product quantified spectrophotometrically by measuring the absorbance at 570 nm wavelength using microplate reader (BioTek ELx800, Highland Park, Winooski, Vermont, USA). The results were presented as a percentage in viabity growth inhibition of treated cells versus cells grown in control medium (indicated as 100%).

2.8. Cytotoxicity assessment – LDH assay

Neurons cultured for 7 days were exposed to serial dilutions of tested fractions (10, 25, 50 and 100 μg/ml) used alone or in combination with 1 mM glutamate, 0.5 mM NMDA, 0.25 mM AMPA. Solutions of investigated compounds were prepared as described above. After 48 h of cells exposure to investigated compounds the culture supernatants were collected in new 96-well microplates, which were used to perform LDH assay following manufacturer's instruction (In vitro Toxicology Assay Kit Lactate Dehydrogenase Based). Absorbance was recorded on a microplate reader (BioTek ELx800) at 450 nm wavelength. The results were presented as a percentage in LDH release to the medium by treated cells versus cells grown in control medium (indicated as 100%).

2.9. Neurite outgrowth assessment

Neurons cultured for 7 days were exposed to serial dilutions of tested fractions (10, 25, 50 and 100 μg/ml) used alone or in combination with 1 mM glutamate, 0.5 mM NMDA, 0.25 mM AMPA. Solutions of investigated compounds were prepared as described above. After 48 h of cells exposure to investigated compounds the neurite outgrowth was assessed using Neurite Outgrowth Staining Kit (Life Technologies) manufacturer's instruction. Briefly, after cell medium removing cells were incubated with Fix/Stain Solution (20 min, 37 °C). Afterwards stain was discarded and cells were exposed to Background Suppression Dye and then neurite outgrowth was analyzed using FLUOstar Omega Microplate Reader (BMG LABTECH) with excitation filter 530 nm; emission filter 590 nm. The results were presented as a percentage in neurite outgrowth of treated cells versus cells grown in control medium (indicated as 100%).

2.10. Antioxidant assessment – ABTS assay

Antioxidant capacity of *C. cibarius* fractions was examined using Antioxidant Assay Kit according to manufacturer's instruction (Sigma-Aldrich). Briefly, myoglobin working solution (20 μl) as well as ABTS (2,2'-azino-bis(3-ethylbenzthiazoline-6-sulfonic acid) working solution (150 μl) prepared by mixing 3% hydrogen peroxide solution (25 μl) and ABTS solution (10 ml) were added to tested fractions (10 μg/ml) and various concentrations of Trolox (standard). After 5 min incubation stop solution (100 μl) was added and absorbance was recorded on a microplate reader (BioTek ELx800) at 405 nm wavelength. The antioxidant activity of each extract was determined by comparing its absorbance with standard (Trolox). The results were presented as TEAC (Trolox Equivalent Antioxidant Capacity) - concentration of substances exhibiting antioxidant activity corresponding to 1 mM Trolox.

2.11. Reactive oxygen species measurements – DCFDA assay

Antioxidant capacity of *C. cibarius* fractions was examined using DCFDA Cellular ROS detection Assay Kit following manufacturer's instruction (abcam, Cambridge, UK). Briefly, 7 days neurons were washed with HBSS (Hank's Balanced Salt Solution) and then exposed to freshly prepared 25 μM DCFDA (2',7'-dichlorofluorescin diacetate) for 45 min at 37 °C. and 50 mM glutamate. Stained cells were washed with PBS and then exposed for 4 h at 37 °C to serial dilutions of fractions CC2 and CC5 (10, 25, 50, 100 μg/ml) applied with or without 55 μM TBHP (Tert-Butyl Hydrogen Peroxide). Amount of fluorescent compound DCF (2',7'-dichlorofluorescei) was measured using FLUOstar Omega Microplate Reader (BMG LABTECH) with excitation filter 485 nm and emission filter 535 nm. Results were expressed as percentage in fluorescent intensity (DCF level) of treated cells versus cells grown in control medium (indicated as 100%).

2.12. Statistical analysis

The data were presented as the mean value and standard error of the mean (SEM). Statistical analysis was performed using the one way-ANOVA with Tukey post-hoc test and column statistics were used for comparisons. Significance was accepted at $p < 0.05$.

3. Results and discussion

According to our best knowledge there are only one evidence which directly suggested neuroprotective properties of *Cantharellus cibarius* extracts and compounds obtained from this mushroom. Orhan and Ustun point out the possibility of use *C. cibarius* methanolic extracts rich in polyphenols in prevention of Alzheimer disease because of proved acetyl cholinesterase inhibition (Orhan & Ustun, 2011). Nevertheless, since inflammatory processes have been closely linked with multiple neurodegenerative pathways (Chen, Zhang, & Huang, 2016) phytochemicals with anti-inflammatory properties occurring in *C. cibarius* could be successfully used for neuroprotection. So far anti-inflammatory actions were proved in case of *C. cibarius* methanolic, ethanolic and water extracts (Moro et al., 2012; Vamanu & Nita, 2014), acetylenic acid derivatives isolated from the methanolic extract of C. cibarius (Hong et al., 2012), 7-dehydrostigmasterol, β-sitosterol, cerebroside, cerevisterol, ergosterol, ergosterol peroxide, glucoside and tuberoside isolated from the n-hexane and ethyl acetate extracts of *C. cibarius* (Kim, Tay, & de Blanco, 2008).

Despite the above mentioned data there are lack of evidences which have shown direct impact of *C. cibarius* and its compounds on nervous system or its function in animals or at least on the neuronal cells. This study is the first attempt to evaluation the neuroprotective activity of polysaccharide fractions isolated from *C. cibarius* in different *in vitro* models of neurodegeneration.

Table 1
Yield and sugar composition of AIR, WSM, WSB and fractions extracted and purified from *Cantharellus cibarius* mushroom.

	Yield (%)	Yield (%)	Fuc	GlcN	Gal	Glc	Man	Rib	Total
AIR	88	88[a]	1.6 ± 0.7	3.0 ± 0.2	tr.	27.0 ± 1.0	7.8 ± 0.8	–	39.4 ± 2.3
WSM	26	29[a]	1.4 ± 0.2	0.1 ± 0.1	0.1 ± 0.1	29 ± 1.7	4.4 ± 0.2	–	35.0 ± 1.1
WSB	1.4	5.4[a]	2.8 ± 2.9	2.7 ± 0.6	3.7 ± 0.3	21.0 ± 2.8	19.0 ± 1.7	2.9 ± 0.3	52.1 ± 1.4
CC1		48[b]	4.4 ± 0.1	tr.	4.0 ± 0.2	9.6 ± 2.2	18.9 ± 3.5	tr.	36.9 ± 5.9
CC2a		6.0[b]	0. 8 ± 1.1	5.0 ± 0.8	4.6 ± 0.2	19.5 ± 2.3	39.3 ± 2.5	tr.	69.2 ± 6.9
CC2b		10.0[b]	5.0 ± 0.1	tr.	4.3 ± 0.1	39.0 ± 0.4	4.9 ± 0.2	tr.	53.2 ± 0.6
CC3		3.1[b]	–	–	–	1.8 ± 0.2	0.7 ± 0.1	8.3 ± 0.8	10.7 ± 1.0

tr. traces; – not detected.
[a] In relation to the mushroom.
[b] In relation to the WSB.

3.1. Sugar composition of investigated fractions

In Table 1 there is shown the yield and sugar composition of the AIR, WSM and WSB of *C. cibarius* mushrooms. As can be observed the WSB corresponds to 1.4% of the original mushroom and to 5.4% of the water soluble material insoluble in alcohol. The WSB was composed by 52.1% of carbohydrates, with glucose and mannose being the most abundant sugars present (40% and 36% w/w, respectively) followed by galactose (7% w/w) being present lower amounts of fucose, glucosamine and ribose (Table 1). The WSB was further fractionated by anion exchange chromatography on a Q-Sepharose FF. As can be observed in Table 1 and Fig. 1 the majority of the material applied was not retained on the column (48%). The retained material and eluted with increasing concentration of NaCl corresponded to 19.3% of the applied material, yielding three fractions CC2a, CC2b and CC3. The overall recovery obtained after anion exchange chromatography was 67.3% of the applied material. As can be observed in Table 1 fractions CC2a and CC2b presented a higher abundance of carbohydrates (69% and 53%, respectively), higher than that observed for CC1 and especially CC3. Fraction CC2a and CC2b presented a different sugar composition, while in CC2a mannose was the main sugar residue (61% w/w) for CC2b glucose residues represented 74% (w/w) of the carbohydrates. For fraction CC3, ribose was the main sugar (77% w/w), and probably correspond to a RNA fraction as observed for *Boletus edulis* mushroom (Lemieszek et al., 2016). In order to evaluate the homogeneity of the obtained polysaccharide fractions, CC1, CC2a and CC2b fractions, the most abundant and polysaccharide enriched fractions, were analyzed by size exclusion chromatography using Sephacryl S-400 HR (molecular weight fractionation range 10 kDa–2000 kDa).

As can be observed in Fig. 2, fraction CC1, not retained in the anion-exchange column was composed by different polysaccharide fractions presenting different molecular weights. On the other hand fractions CC2a and CC2b presented a more homogenous profile, being clearly observed a main carbohydrate peak closely followed by the absorbance at 280 nm, suggesting the presence of associated proteins. In order to have a deeper insight into the structural details of the polysaccharides present in each fraction and as fraction CC2a and CC2b presented a more homogeneous structure, the fractions CC1SEA, CC1SEB and CC1SEC obtained after size exclusion chromatography and the fractions CC2a and CC2b were analyzed by methylation analysis (Table 2).

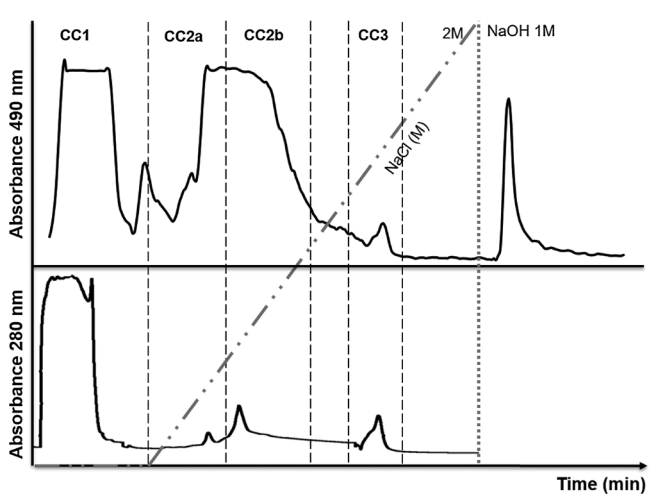

Fig. 1. Purification of the *C. cibarius* WSB by anion-exchange chromatography on Q-Sepharose FF.

As can be observed in Table 2, for all fractions analyzed the mannose residues were mainly linked by 1,6 and 1,4 linkages, although there were observed differences for the different fractions with CC1SEA, CC2a and CC2b presenting an higher abundance of 1,6-linked mannose residues and fractions CC1SEB and CC1SEC presenting an higher abundance of 1,4 linked mannose residues (Table 2). In all fractions analyzed there were also observed high abundances of 1,2,6 and 1,3,6-linked mannose residues, suggesting that these mannans were highly branched, nevertheless the branching degree and also the branching pattern observed for the different mannans were different (Table 2). Mannans retained in the anion-exchange column presented a higher branching degree compared to those not retained (BD, Table 2) and additionally for the retained mannans there was observed a higher abundance of 2,6-linked mannose residues in comparison to the 3,6-linked mannose residues in relation to those mannans not retained (Table 2). Glucose residues were mainly 1,3-linked for the polysaccharide fractions retained in the anion-exchange column, on the contrary for fractions CC1SEB and CC1SEB, glucose residues were mainly 2,3-linked, showing a more branched structure of the glucose residues in these fractions. Fucose residues were mainly terminally linked and galactose residues were mainly substituted in the *O*-2 in all fractions analyzed.

The structural details of the polysaccharides isolated and purified in this work are partially in accordance with the structural details recently obtained for polysaccharides and N-linked glycoproteins isolated from *C. cibarius*. This mushroom has previously been found to contain a 150 kDa water soluble (1,6)-α-glucan with α-(1,4)-linked side chains (Villares, García-Lafuente, Guillamón, & Mateo-Vivaracho, 2013). Han et al. reported a protein-bound polysaccharide containing an (1,6)-linked glucan with single glucose residues linked to *O*-3 (Han, Li et al., 2013). Nyman et al. isolated from fruiting bodies of *C. cibarius* a water soluble (1,6)-linked α-D-mannan substituted with single and short (1,2)-linked α-D-mannan side chains and two types of β-glucans: a β-D-(1,6)-linked backbone containing single glucose and β-(1,3)-linked glucan side chains and a less branched (1,3)-linked β-D-glucan with side chains in *O*-4 that may contain (1,4)-linked glucose in addition to a terminal xylose residue (Nyman, Aachmann, Rise, Balance, & Samuelsen, 2016). Grass et al isolated N-glycans from *C. cibarius* that contained terminally linked L-fucose in α-1,6-linkage to α-1,6-mannose of oligomannosidic N-glycans as well as to the first GlcNAc. Besides fucosylated N-glycans, short *O*-linked mannan chains substituted with fucose were present on

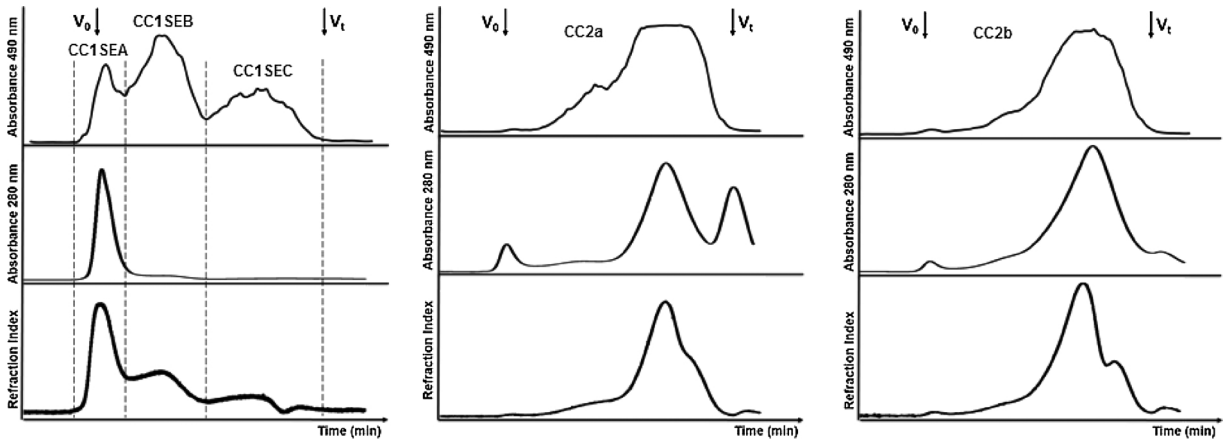

Fig. 2. Size exclusion chromatograms of polysaccharide enriched fractions CC1, CC2a and CC2b on Sephacryl S-400 HR, monitored continuously by refraction index and UV-detection at 280 nm, and collected fractions (2 ml) were analyzed for the total carbohydrate content by the Phenol-sulphuric acid method (abs 490 nm).

Table 2
Linkage profile of the water soluble ethanol insoluble polysaccharides of *C. cibarius* recovered in different fractions.

Linkages	CC1SEA	CC1SEB	CC1SEC	CC2a	CC2b
T-Fuc*p*	4.2	4.9	8.5	2.6	8.1
2-Fuc*p*	0.2	0.1	–	–	0.5
Total	4.4	5.0	8.5	2.6	8.6
T-Man*p*	20.7	20.8	18.5	16.0	12.1
3-Man*p*	0.3	0.3	0.4	1.7	1.1
4-Man*p*	12.3	20.1	31.5	4.0	7.2
6-Man*p*	35.2	23.7	11.8	36.8	23.6
4,6-Man*p*	2.9	–	–	–	–
2,6-Man*p*	8.9	9.4	12.9	16.5	12.5
3,6-Man*p*	6.6	7.0	1.5	1.3	3.3
Total	86.8	81.3	76.4	76.2	59.8
BD[a]	21.2	20.2	18.8	23.4	26.4
2,6-Man*p*/3,6-Man*p*	1.35	1.34	8.60	12.7	3.79
%T-Man*p*/Total Man*p*	23.8	25.6	24.2	21.0	20.2
T-Gal*p*	0.6	0.7	0.9	0.8	0.5
2-Gal*p*	5.3	5.7	6.0	4.7	4.0
3-Gal*p*	1.5	0.6	–	–	
Total	7.5	7.0	6.9	5.5	4.5
T-Glc*p*	0.1	0.4	0.4	2.0	1.4
3-Glc*p*	–	0.2	0.2	13.6	25.7
4-Glc*p*	1.2	0.2	–	–	–
6-Glc*p*	–	1.0	0.7	–	–
3,4-Glc*p*	–	–	0.6	–	–
2,3-Glc*p*	–	5.0	6.3	–	–
3,6-Glc*p*	–	0.4	0.6	–	–
4,6-Glc*p*	–	3.0	–	–	–
Total	1.3	10.1	14.8	15.6	27.1

[a] BD – branching degree calculated as percentage of 2,3-Man*p* + 3,6-Man*p* + 4,6-Man*p*/Total Man*p*.

chanterelle proteins (Grass et al., 2011). Chen et al. isolated and purified a polysaccharide from the fruiting body of *C. cibarius* composed by L-arabinose, D-mannose, D-glucose, and D-galactose in a molar ratio of 1:1.06:1.95:1.17 with a molecular weight of 336 kDa (Chen et al., 2017). Variations in polysaccharide structures reported from the same mushroom may be due to differences in extraction procedures and purification techniques used. The structural features of the polysaccharides isolated and purified in this work have more similarities to those reported by Nyman et al. concerning the structures of mannan and glucan residues, nevertheless these authors didn't described the presence of terminally linked fucose residues and glucosamine residues as found in our work, nevertheless these structural details being in accordance with the results of Grass et al. (2011), Nyman et al. (2016). Also the presence of galactose residues in *C. cibarius* polysaccharides was not described previously with the exception of the work by Chen et al. nevertheless in this last work no structural details about the galactose linkage pattern was presented (Chen et al., 2017).

3.2. *C. cibarius* fractions influence on neurons viability and neurite outgrowth

In order to define *C. cibarius* polysaccharides neuroprotective abilities compounds influence on neurons viabilty was assessed by MTT and LDH tests. Furthermore neurite outgrowth was also examined. As presented on Fig. 3A, MTT test revealed that fractions CC2a and CC3 at the concentration 100 μg/ml increased metabolic activity of neurons (average of about 10%), while fractions CC2b did not affected mitochondrial activity in investigated cells. On the contrary fraction CC1 at concentration 100 μg/ml significantly decrease neurons viability by around 30%. The MTT data correspond with results obtained from LDH assay (Fig. 3B), which revealed damage in the cell membrane of neurons treated with CC1 a sign of what was the increase of lactate dehydrogenase (LDH) concentration in cell culture medium average about 20%. Instead other investigated fractions were not toxic against neurones. Furthermore, fraction CC2a (100 μg/ml) and CC3 (50 μg/ml, 100 μg/ml) significantly decreased LDH concentration in medium collected from neurons cultures, which may suggested trophic properties of mentioned fraction. The results described above corresponded with data obtained from assessment of neurite outgrowth (Fig. 3C). Once again fractions CC2a and CC3 revealed beneficial impact on neurite outhgrowt, while fractions CC2b did not affected on the tested process. On the contrary fractions CC1 significantly decreased neurite outhgrowt.

Among several studies, which have shown beneficial impact of mushroom compounds on both neuronal cells viability and neurite growth only two conducted by Park et al. and Kim et al. concerned polysaccharides (Kim et al., 2007; Park et al., 2002; Phan, David, Naidu, Wong, & Sabaratnam, 2013; Rahman, Abdullah, & Aminudin, 2016). Studies performed by Park et al. on rat pheochromocytoma nerve cell line PC12 revealed that exopolysaccharide (composed of glucose:galactose:xylose:mannose:fructose; molar ratio of 1.5:1.7:1.2:0.6:0.9) isolated from *Hericium erinaceus* enhanced viability of nerve cells and improved the extension of neurites (Park et al., 2002). Kim et al. have shown promotion of neuritogenesis in subclone of PC12 cells (PC12 h) by *Tremella fuciformis* water extract (comprised high polysaccharides including β(1 → 3)-glucans acidic heteroglycans and glucuroxylomannan) and furthermore revealed neutralization of scopolamine-induced deficit in learning and memory in rats (Kim et al., 2007). Several promising results were obtained in research on *Ganoderma lucidum* polysaccharides (GLPS). Zhao et al. reveled that GLPS

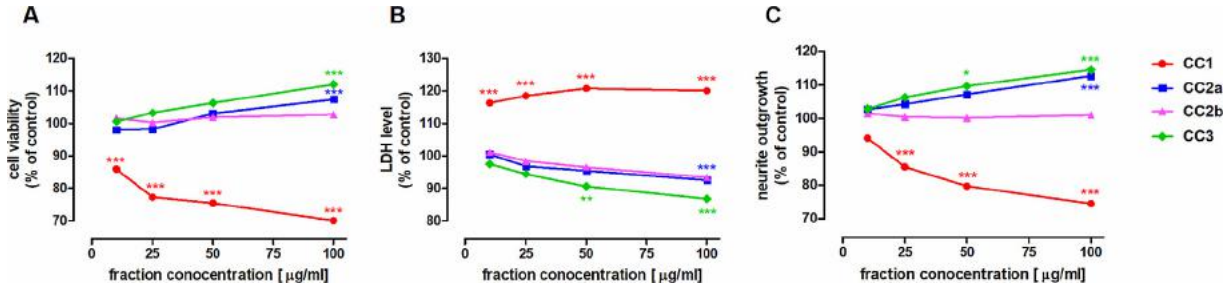

Fig. 3. Influence of *Cantharellus cibarius* fractions (CC1–CC3) on viability of neurons derived from human neuroblastoma cell line SH-SY5Y as well as neurite outgrowth. Cells were exposed to culture medium alone (control) or tested fractions in concentrations ranging from 10 to 100 μg/ml for 48 h. Cell viability was measured photometrically by MTT assay (A) and LDH assay (B). Growth of neuritis was measured after fluorescence staining using Neurite Outgrowth Staining Kit (C). Results are presented as the mean of 3–6 measurements. * $p < 0.05$ *vs*. control, ** $p < 0.01$ *vs*. control, *** $p < 0.001$ *vs*. control, one-way ANOVA test; post test: Tukey.

(investigated polysaccharides consisted of rhamnose, xylose, fructose, galactose, mannose, and glucose with molar ratios of 0.793 : 0.964 : 2.944 : 0.167 : 0.384 : 7.94 and were linked by β-glycosidic linkages) increased viability in cerebral cortical neurons exposed to ischemia/reperfusion (Zhao et al., 2004). This observation was confirmed by Zhou et al. after aplication water-soluble GLPS (the major carbohydrate found in tested polysaccharides were galactose and glucose; molar ration of 1:12.3) (Zhou et al., 2010). Furthermore, Seow et al. reveled that crude polysaccharides of *Lignosus rhinocerotis* sclerotium stimulated neuritogenesis in PC12 cells. Unfortunatly the chemical nature of investigated compound was not defined (Seow et al., 2015).

Because of chemical nature of CC3 fraction (RNA fraction) studies regarding cordycepin (3'-deoxyadenosine isolated from *Cordyceps militaris*) should be mentioned. They revealed that cordicepin in low concentrations did not affected viability of murine hippocampal neuronal cell line HT22 (Jin et al., 2014), murine microglial cell line BV2 and murine hippocampal neurons (Peng, Wang, Ge, Qu, & Jin, 2015). Based on obtained results from further studies were selected fractions CC2a and CC3, which revealed beneficial impact on neurons viability as well as function (neurite outgrowth). Attention is drawn to the fact that the best results were obtained in case of fractions with the largest (CC2a – 69.2%) and smallest (CC3 – 10.7%) sugar content (Table 1).

3.3. Neuroprotective effect of C. Cibarius fractions under trophic stress conditions

Trophic deprivation mediated neuronal death is important during nervous system development (Oppenheim, 1991; Purves, Snider, & Voyvodic, 1988) on the other hand in the formed as well as developed nervous system leads to neurodegeneration (Sheline, Cai, Zhu, & Shi, 2010). Because of that the use of neurotrophic factors such as nerve growth factor (NGF), neurotrophin 3 (NT-3), glia-derived neurotrophic factor (GDNF), brain-derived neurotrophic factor (BDNF) as therapeutic agents is a novel approach aimed at maintaining and even restoring neuronal function (Connor & Dragunow, 1998). On the other hand there is widespread interest in the discovery molecules able to prevent and/or reduce the neuronal cell death observed in neurodegenerative diseases and/or enhancing the action of endogenous neurotrophins.

In order to examine the influence of selected fractions on neurons viabilty under trophic stress conditions as well as define their nutritive potential, trophic factor (B27) was removed from the medium used for neuron culture. Because of the fact that absence B27 did not damaged cell membrane integrity but influence on cell viability as well as neurite outgrowth studies were performed using only MTT test and Neurite Outgrowth Staining Kit, respectively. As presented on Fig. 4A, B27 supplement deprivation (SD) decreased mitochondrial activity of neurons by more than 24% compared to control. Furthermore, MTT assay demonstrated that fraction CC2a did not affect the viability of the neurons under the analyzed stress conditions, on the contrary fraction CC3 at the concentration 50 and 100 μg/ml neutralized B27 deficiency and maintained and/or restored mitochondrial dehydrogenase activity on the level observed in neurons growing in the medium containing B27. As shown on Fig. 4B, absence B27 inhibited growth of neurites by approximately 25% compared to control conditions. Fraction CC2a did not affected neurite outgrowth under trophic stress conditions on the contrary fraction CC3 applied in the medium without B27 supplement effectively neutralized serum deficiency. Obtained results suggested that fraction CC3 contains some nutritive compounds, which are able to replace trophic factors presented in B27 supplement. The similar impact on neuronal cells viability and neurite outgrowth was reported by previously mentioned Park et al. studies, who revealed that *Hericium erinaceus* exopolysaccharide better influence on the analyzed parameters in rat pheochromocytoma nerve cell line PC12 than tested growth factors: NGF and BDNF (Park et al., 2002). On the contrary cited earlier Kim et al. studies suggested that promotion of neuritogenesis in subclone of PC12 cells (PC12h) by *Tremella fuciformis* polysaccharides is a results of NGF stimulation (Kim et al., 2007). Nevertheless neurotrophic abilities of fraction CC3 could be also connected with RNA presence, which correspond with Koike data, who have shown that cordycepin inhibited degenerative processes of PC12 cells after acute withdrawal of NGF (Koike, 1992).

3.4. Neuroprotective effect of selected C. cibarius fractions in excitotoxicity model of neurodegeneration

Excitotoxicity is the pathological process wherein excessive stimulation of the glutamate receptors by increased concentrations of their agonist, or their long-term activation by moderately growing concentrations of the excitatory amino acids, leads to nerve cells damage or death (Lucas & Newhouse, 1957). The principal excitatory neurotransmitter in the central nervous system (CNS) is glutamate (Arundine & Tymianski, 2004), which acts through two major classes of receptors: glutamate-gated ion channels (ionotropicglutamate receptors, iGluRs) known as AMPA (α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid), NMDA (*N*-methyl-D aspartate) and kainate receptors and G protein-coupled metabotropic receptors (mGluRs) (Panter, Yum, & Faden, 1990; Schmidt, Bubser, & Hauber, 1990). Understanding of the mechanism of excitotoxicity revealed that neuronal injury is mainly mediated by NMDA receptors (Zhou et al., 2015). Nevertheless AMPA receptors are also involved in glutamate evoked neurodegeneration but indirectly (Doble, 1999). In order to examined neuroprotective abilities of selected fractions (CC2a and CC3) in excitotoxicity model of neurodegeneration, their influence on neurons viability and their growth were assessed in presence of glutamatergic system activators: 1 mM glutamate, 0.5 mM NMDA and 0.25 mM AMPA. Studies were performed using MTT and LDH tests as well as Neurite Outgrowth Staining. Inducers of glutamate receptors were used in concentrations which decreased cells viability by around 25% (Fig. 5). Investigated fractions

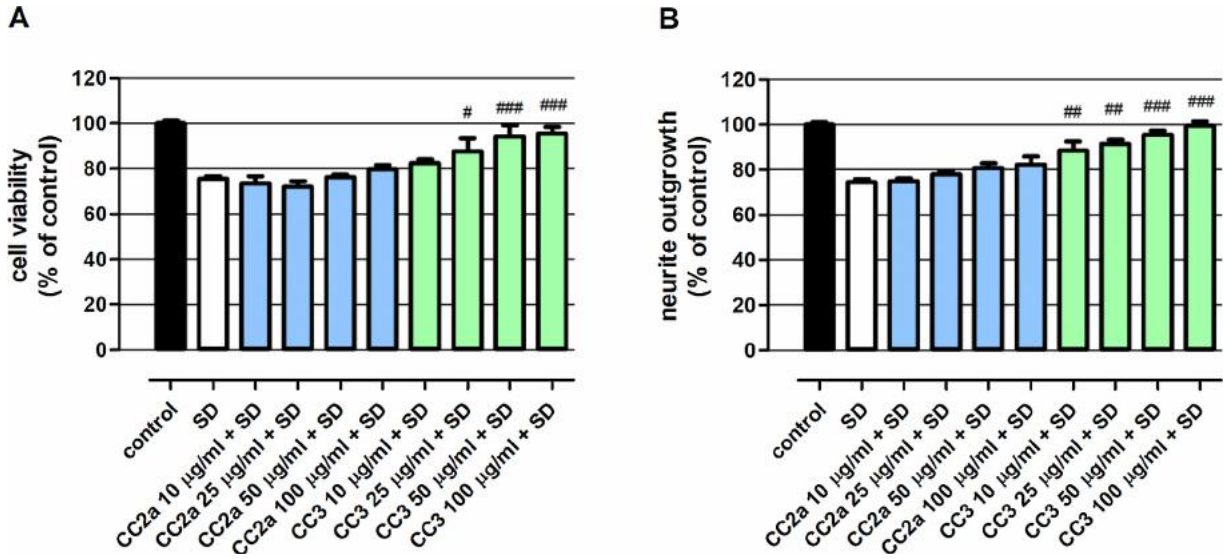

Fig. 4. Neuroprotective effect of *Cantharellus cibarius* fractions (CC2a, CC3) under trophic stress conditions in neurons derived from human neuroblastoma cell line SH-SY5Y. Neurotoxicity was induced by B27 supplement deprivation. Cells were exposed to culture medium with (control) and without B27 (SD) or tested fractions in concentrations ranging from 10 to 100 for 48 h. Cell viability was measured photometrically by MTT assay (A). Growth of neuritis was measured after fluorescence staining using Neurite Outgrowth Staining Kit (B). Results are presented as the mean ± SEM of 3–6 measurements. # $p < 0.05$ *vs*. SD, ## $p < 0.01$ *vs*. SD, ### $p < 0.001$ *vs*. SD, one-way ANOVA test; post test: Tukey.

effectively eliminated excitotoxicity induced by abovementioned compounds. Performed studies revealed statistically significant increase of mitochondrial dehydrogenase activity (Fig. 5A, D, G), decrease of lactate dehydrogenase activity (Fig. 5B, E, H) as well as restoration neurite outgrowth (Fig. 5C, F, I) after cells exposure to tested fractions in higher concentrations (50 and 100 µg/ml) applied together with glutamate or NMDA or AMPA. The strongest neuroprotective effect was observed in case of fraction CC3, which used at concentration 100 µg/ml completely eliminated the negative effects of glutamatergic system activators. Presented studies correspond with data recorded by Hu et al., who revealed neuroprotective effects of purified *Sparassis crispa* polysaccharides against glutamate-induced toxicity in differentiated PC12 cells (Hu et al., 2016). Similar results in the same *in vitro* model were obtained by Jin et al. in case of *Tremella fuciformis* polysaccharides (Jin et al., 2016). Olatunji et al. also revealed neuroprotective properties of polysaccharides purified from *Cordyceps cicadae* against glutamate-induced damage in PC12 cells (Olatunji et al., 2016). Because of chemical nature of CC3 fraction (RNA fraction) two other studies regarding cordycepin (3′-deoxyadenosine isolated from *Cordyceps militaris*) should be mentioned. Jin et al. explored neuroprotective effects of cordycepin against glutamate-induced cytotoxicity in murine hippocampal neuronal cell line HT22, which was associated with an adenosine receptor–dependent pathway (Jin et al., 2014). On the contrary Cheng et al. have shown that cordycepin was able to exert a potent neuroprotective function in ischemic mice by reduction of extracellular levels of glutamate and aspartate (Cheng et al., 2011).

3.5. Neuroprotective effect of selected C. cibarius fractions under oxidative stress conditions

Accumulating evidence have indicated that oxidative stress has been implicated in several degenerative processes including neurodegenerative diseases (Liu, Zhou, Ziegler, Dimitrion, & Zuo, 2017). Particularly vulnerable to oxidative damage are neurons because of their high oxygen consumption, high content of polyunsaturated fatty acid in membranes and low efficiency of antioxidant systems (Rego & Oliveira, 2003). Therefore, a compounds with antioxidant activity which may prevent neuronal degeneration are of special interests. The total antioxidant capacity (TAC) of selected fractions isolated from *C. cibarius* was examined using commercial Antioxidant Assay Kit, which allows to compare TAC of investigated compounds with Trolox, a water-soluble vitamin E analog. As presented on Fig. 6A, 10 µg/ml solutions of selected fractions have shown antioxidant activity corresponding to 1.278 mM (CC2a) and 0.412 mM (CC3) Trolox. Observed antioxidant properties of tested fractions repeatedly exceed ABTS + radical-scavenging capacity of polysaccharide extracts isolated from *Auricularia auricul*, *Dictyophora indusiate*, *Hericium erinaceus*, *Hohenbuehelia serotina*, *Hypsizygus marmoreus*, *Lentinus edodes, Pleurotus eryngii Quel, Russula vinosa Lindblad*. Antioxidant capacities of above mentioned extracts ranged from 1.70 ± 0.42 to 65.98 ± 1.74 µM TE/g crude polysaccharide (Li, Wang, Wang, Walid, & Zhang, 2012). Moreover studies performed by Li et al. reveled a positive correlation between amount of crude polysaccharides with total antioxidant activity which correspond with our observation. Pronounced antioxidant capacity of *C. cibarius* polysaccharides were also reported by Zhao et al., who have shown CC-1 (MW: 61.056 kDa; CC-1 was mainly formed of the glucose and xylose at ratio of 5:1; backbone of 1,4-linked-β-D-glucose which branched at *O*-6 and the branches were mainly composed of 6 → 1)-α-D-xylopyranose residue) ability to clear the ABTS + free radical and DPPH- free radical in a certain range of concentration (0.1–5 mg/ml) (Zhao et al., 2018). Nevertheless most of the existing data assigned free radicals scavenging ability to *C. cibarius* flavonoids and phenolic compounds (Ebrahimzadeh et al., 2010, 2015; Kozarski et al., 2015; Palacios et al., 2011; Vamanu & Nita, 2014; Witkowska, Zujko, & Mirończuk-Chodakowska, 2011).

In the next step neuroprotective effect of tested fractions were examined under oxidative stress conditions induced in neuron cells by 50 µM TBHP. In order to examine the fractions ability of neutralization of reactive oxygen species (ROS) DCFCA Cellular ROS Detection Assay Kit was applied (Fig. 6B). Nevertheless at the beginning fractions impact on exogenous level of ROS was investigated. Performed studies revealed that both CC2a and CC3 in full range of analyzed concentrations did not generated ROS in neuronal cells At the same time tested fractions decrease the amount of DCF (marker of oxidative stress) induced by TBHP. The most effective free radical scavenger turned out to be fraction CC2a, which reduced DCF level by 19.2% (10 µg/ml), 30.0% (25 µg/ml), 53.3% (50 µg/ml) and 69.3% (100 µg/ml) compared to cells exposed to 50 µM TBHP. Fraction CC3 which revealed three times

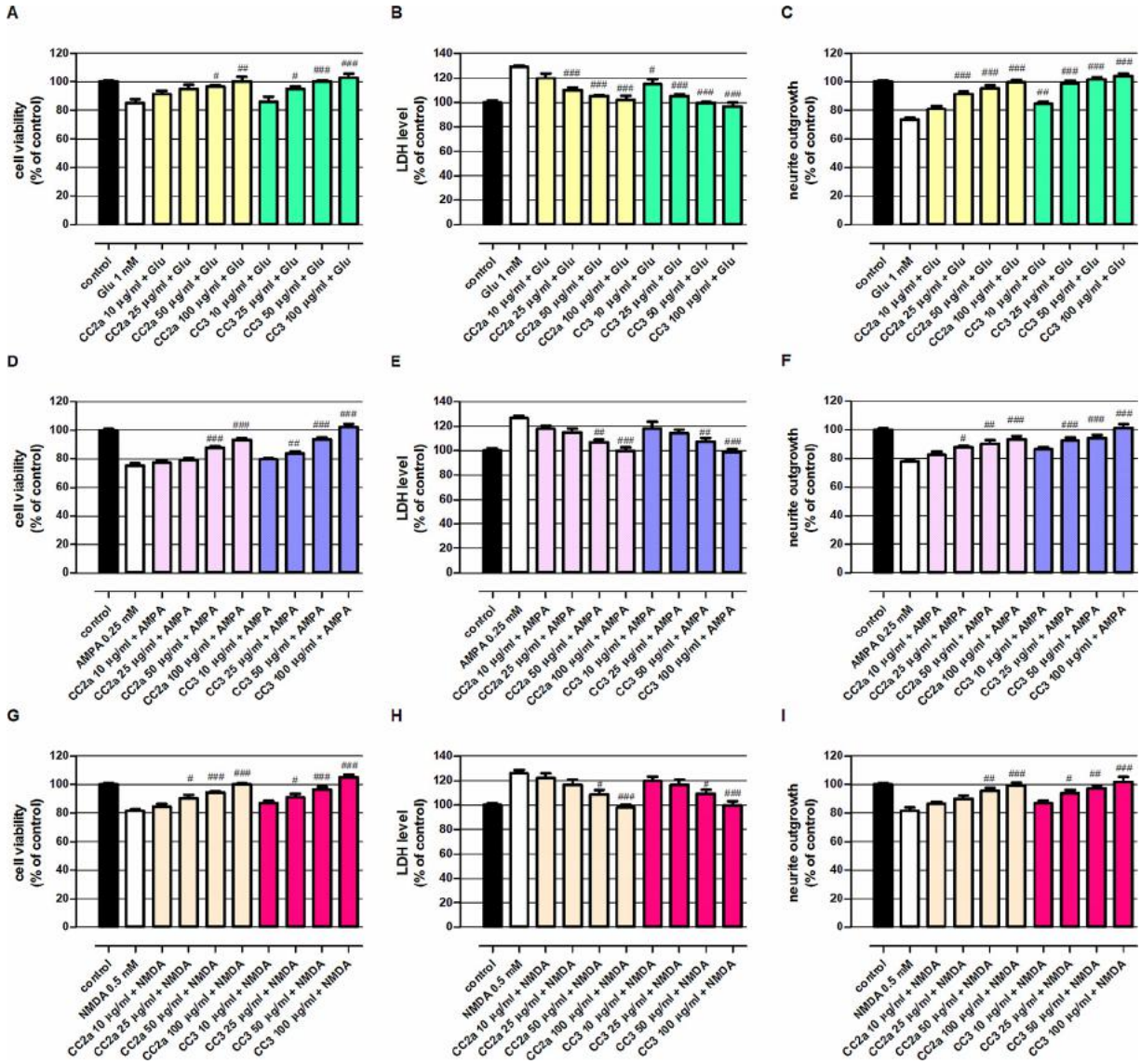

Fig. 5. Neuroprotective effect of *Cantharellus cibarius* fractions (CC2a, CC3) against excitotoxicity induced in neurons derived from human neuroblastoma cell line SH-SY5Y. Excitotoxicity was induced by neurons exposure to 1 mM glutamate (Glu), 0.5 mM NMDA and 0.25 mM AMPA. Cells were exposed for 48 h to culture medium alone (control) or medium containing above-mentioned compounds as well as tested fractions in concentrations ranging from 10 to 100. Cell viability was measured photometrically by MTT assay (A, D, G) and LDH assay (B, E, H). Growth of neuritis was measured after fluorescence staining using Neurite Outgrowth Staining Kit (C, F, I). Results are presented as the mean ± SEM of 3–6 measurements. # $p < 0.05$ *vs*. glutamate/NMDA/AMPA, ## $p < 0.01$ *vs*. glutamate/NMDA/AMPA, ### $p < 0.001$ *vs*. glutamate/NMDA/AMPA, one-way ANOVA test; post test: Tukey.

lower TAC than CC3, significantly reduced DCF level (by 15.3%) only in the highest tested concentration (100 µg/ml). Similar to results of ABTS assay cell based test also revealed higher antioxidant capacity of fraction CC2a than CC3, which can be linked with high content of polysaccharides (69.2% *vs*. 10.7%). It needs to be highlighted that among several studies on antioxidant properties of compound isolated from *C. cibarius* there is lack of evidences performed on cells, which raises the importance of our research (Ebrahimzadeh et al., 2010, 2015; Kozarski et al., 2015; Palacios et al., 2011; Vamanu & Nita, 2014; Witkowska et al., 2011; Zavastin et al., 2016). This issue is not well researched also in the context of research on mushroom polysaccharides at all. So far neuroprotective effect of mushroom polysaccharides under oxidative stress conditions were reported only in case of high molecular weight polysaccharides isolated from *Hericium erinaceus* (170 kDa and 110 kDa) (Cheng et al., 2016).

4. Conclusion

Summarizing our studies revealed beneficial effect of *Cantharellus cibarius* fractions (CC2a, CC3) on neurons viability as well as neurite outgrowth in both normal and different stress conditions. Both tested fractions have shown antioxidant capacity as well as effectively neutralize the negative changes induced by activators of glutamatergic system (glutamate, NMDA, AMPA). Furthermore performed studies revealed that fraction CC3 acted as a trophic factor. Discovered neuroprotective properties of investigated compounds suggested the their use for developing effective and safety therapeutic strategy for neurodegenerative diseases.

Acknowledgments

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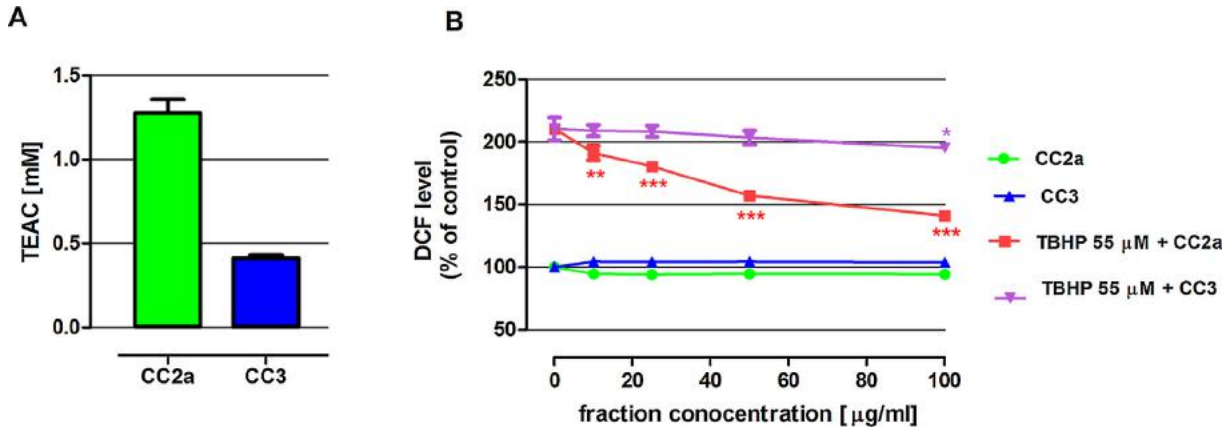

Fig. 6. Antioxidant properties of *Cantharellus cibarius* fractions (CC2a, CC3). A) Total antioxidant capacity of fractions CC2a and CC3. Investigated fraction were used in concentration 10 μg/ml. Results are presented as the mean ± SEM of 8 measurements. TEAC (Trolox Equivalent Antioxidant Capacity) - concentration of substances exhibiting antioxidant activity corresponding to 1 mM Trolox. B) Neuroprotective effect of fractions CC2a and CC3 under oxidative stress conditions induced in 7 days neurons by 55 μM TBHP. Cells were preincubated with DCFDA for 45 min. and afterwards exposed for 4 h to investigated fractions in concentrations ranging from 10 to 100 μg/ml with and without TBHP. Antioxidant capacity of tested extract was determined by measure of fluorescence intensity of DCF - product of DCFDA oxidation. Results are presented as the mean ± SEM of 3 measurements. * $p < 0.05$ *vs*. TBHP, ** $p < 0.01$ *vs*. TBHP, *** $p < 0.001$ *vs*. TBHP, one-way ANOVA test; post test: Tukey.

cibarius (golden chanterelle) under oxidative stress conditions - *in vitro* studies"; Project Number 14050 Evaluation of the neuroprotective properties of polysaccharide fractions obtained from *Cantharellus cibarius* (golden chanterelle) in the human neuron culture derived from cell line SH-SY5Y". We appreciate the financial support provided to the Research Unit in Vila Real (PEst-OE/QUI/UI0616/2014) by FCT (Fundação para a Ciência e a Tecnologia) Portugal and COMPETE2020 (Progama Operacional Competitividade e Internacionalização).

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